Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How load does it take to download the file?

Reasoning + Answer:

<think> To solve this problem, let's first calculate how much of the file Carla had downloaded before the Windows update forced a restart. Since the total file size is 200 GB and 40% of the way through the download would be at \(0.40 \times 200 = 80 \) GB, we know she had downloaded 80 GB before the update. Next, we need to determine how much time it took her to download these 80 GB. Given that her download speed is 2 GB per minute, we can find the time by dividing the amount downloaded by her download speed: \(80 \div 2 = 40 \) minutes. So, she spent 40 minutes downloading the initial 80 GB before the update. Now, we know that the update itself took 20 minutes, which means this time was not spent downloading file content but rather on system operations. Therefore, we need to add this 20-minute update time to the initial 40 minutes of download time to get the total time spent so far: \(40 + 20 = 60 \) minutes. Finally, we need to calculate the remaining download time. Since she has already downloaded 80 GB and the total file size is 200 GB, the remaining amount to download is \(200 - 80 = 120 \) GB. Given her download speed of 2 GB per minute, we can find out how many more minutes it will take to download the remaining 120 GB by dividing 120 by 2: \(120 \div 2 = 60 \) minutes. To find the total time required to download the entire file, we add the time spent downloading the initial 80 GB (40 minutes), the time spent on the update (20 minutes), and the time needed to download the remaining 120 GB (60 minutes): \(40 + 20 + 60 = 120 \) minutes. </think><answer> 120 </answer>